

The Gram Matrix

Exact Computation via the Digamma Function

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Independent

Abstract

We compute the Gram matrix $G_{kl} = \langle \rho_{1/k}, \rho_{1/l} \rangle$ of the Beurling–Nyman functions exactly. The key is a simple observation: after the substitution $x = 1/t$, the function $\rho_{1/k}(1/t)$ is piecewise constant, taking the value j/k on each interval $[kn+j, kn+j+1)$. This reduces the computation to sums over intervals, which evaluate to differences of the digamma function. We prove:

$$\|\rho_{1/k}\|^2 = \frac{1}{k^3} \sum_{j=1}^{k-1} j^2 \left(\psi\left(\frac{j+1}{k}\right) - \psi\left(\frac{j}{k}\right) \right),$$

where ψ is the digamma function. For $k = 2$: $\|\rho_{1/2}\|^2 = (\log 2)/4$. For the general inner product, the same method applies with $\text{lcm}(k, l)$ periodicity. The first alignment angle satisfies $\cos^2 \theta_2 = \log 2 \approx 0.693$.

1 The Substitution

Lemma 1 (Piecewise constancy after substitution). *For $k \geq 2$ and $t > 0$, write $t = kn + j + s$ where $n \geq 0$, $j \in \{0, 1, \dots, k-1\}$, and $s \in [0, 1)$. Then:*

$$\rho_{1/k}\left(\frac{1}{t}\right) = \left\{\frac{t}{k}\right\} - \frac{1}{k}\{t\} = \frac{j}{k}.$$

That is, $\rho_{1/k}(1/t) = j/k$ on each interval $[kn + j, kn + j + 1)$.

Proof. For $t \in [kn + j, kn + j + 1)$ with $0 \leq j \leq k-1$: $t/k \in [n + j/k, n + (j+1)/k)$, so $\{t/k\} = t/k - n$. Also $t \in [kn + j, kn + j + 1)$ means $\lfloor t \rfloor = kn + j$, so $\{t\} = t - kn - j$. Therefore:

$$\left\{\frac{t}{k}\right\} - \frac{1}{k}\{t\} = \left(\frac{t}{k} - n\right) - \frac{1}{k}(t - kn - j) = \frac{t}{k} - n - \frac{t}{k} + n + \frac{j}{k} = \frac{j}{k}. \quad \square$$

Remark 2 (Values). $\rho_{1/k}(1/t) = 0$ on $[kn, kn + 1)$ and increases in steps of $1/k$:

$t \in$	$[kn, kn + 1)$	$[kn + 1, kn + 2)$	$\dots$	$[kn + k - 1, kn + k)$
$\rho_{1/k}(1/t)$	0	$1/k$	$\dots$	$(k-1)/k$

The function $\rho_{1/k}(1/t)$ is periodic with period k and piecewise constant with values in $\{0, 1/k, \dots, (k-1)/k\}$.

2 The Norm Formula

Theorem 3 (Exact norm formula). *For $k \geq 2$:*

$$\|\rho_{1/k}\|^2 = \frac{1}{k^3} \sum_{j=1}^{k-1} j^2 \left(\psi\left(\frac{j+1}{k}\right) - \psi\left(\frac{j}{k}\right) \right),$$

where ψ is the digamma function $\psi(z) = \Gamma'(z)/\Gamma(z)$.

Proof. By the substitution $x = 1/t$ ($dx = dt/t^2$):

$$\|\rho_{1/k}\|^2 = \int_0^\infty \rho_{1/k}(x)^2 dx = \int_0^\infty \rho_{1/k}(1/t)^2 \frac{dt}{t^2}.$$

By Lemma 1, $\rho_{1/k}(1/t) = j/k$ on $[kn + j, kn + j + 1)$, so:

$$\|\rho_{1/k}\|^2 = \sum_{j=0}^{k-1} \frac{j^2}{k^2} \sum_{n=0}^\infty \int_{kn+j}^{kn+j+1} \frac{dt}{t^2} = \sum_{j=1}^{k-1} \frac{j^2}{k^2} \sum_{n=0}^\infty \left(\frac{1}{kn+j} - \frac{1}{kn+j+1} \right).$$

The inner sum factors via the digamma formula (for $b > a$: $\sum_{n=0}^\infty 1/((kn+a)(kn+b)) = (\psi(b/k) - \psi(a/k))/(k(b-a))$):

$$\sum_{n=0}^\infty \frac{1}{kn+j} - \frac{1}{kn+j+1} = \sum_{n=0}^\infty \frac{1}{(kn+j)(kn+j+1)} = \frac{\psi((j+1)/k) - \psi(j/k)}{k}.$$

Therefore:

$$\|\rho_{1/k}\|^2 = \sum_{j=1}^{k-1} \frac{j^2}{k^2} \cdot \frac{\psi((j+1)/k) - \psi(j/k)}{k} = \frac{1}{k^3} \sum_{j=1}^{k-1} j^2 \left(\psi\left(\frac{j+1}{k}\right) - \psi\left(\frac{j}{k}\right) \right). \quad \square$$

3 Explicit Values

Proposition 4 (Norm for $k = 2$).

$$\|\rho_{1/2}\|^2 = \frac{\log 2}{4}.$$

Proof. $\|\rho_{1/2}\|^2 = \frac{1}{8}(\psi(1) - \psi(1/2))$. Using $\psi(1) = -\gamma$ and $\psi(1/2) = -\gamma - 2\log 2$ (standard values): $\psi(1) - \psi(1/2) = 2\log 2$. Therefore $\|\rho_{1/2}\|^2 = (2\log 2)/8 = (\log 2)/4$. $\square$

Proposition 5 (Norm for $k = 3$).

$$\|\rho_{1/3}\|^2 = \frac{6\log 3 - \pi/\sqrt{3}}{27}.$$

Proof. $\|\rho_{1/3}\|^2 = \frac{1}{27}[1^2(\psi(2/3) - \psi(1/3)) + 2^2(\psi(1) - \psi(2/3))] = \frac{1}{27}[\psi(2/3) - \psi(1/3) + 4\psi(1) - 4\psi(2/3)] = \frac{1}{27}[4\psi(1) - 3\psi(2/3) - \psi(1/3)]$. Using: $\psi(1) = -\gamma$, $\psi(1/3) = -\gamma - \frac{3}{2}\log 3 - \frac{\pi}{2\sqrt{3}}$, $\psi(2/3) = -\gamma - \frac{3}{2}\log 3 + \frac{\pi}{2\sqrt{3}}$ (from the Gauss digamma theorem and the reflection formula $\psi(2/3) - \psi(1/3) = \pi \cot(\pi/3) = \pi/\sqrt{3}$):

$$4\psi(1) - 3\psi(2/3) - \psi(1/3) = 6\log 3 - \frac{\pi}{\sqrt{3}}. \quad \square$$

Remark 6 (Numerical values). $\|\rho_{1/2}\|^2 = (\log 2)/4 \approx 0.1733$.

$$\|\rho_{1/3}\|^2 = (6\log 3 - \pi/\sqrt{3})/27 \approx (6.592 - 1.814)/27 \approx 0.177.$$

The two norms are nearly equal; the asymptotic $\|\rho_{1/k}\|^2 \sim C/k$ holds only for large k , with $C = \int_0^1 x^2 \psi'(x) dx$.

4 The Gram Matrix

Theorem 7 (Gram matrix formula). *For $k, l \geq 2$ and $m = \text{lcm}(k, l)$, write the period- m pattern:*

$$\rho_{1/k}(1/t) \rho_{1/l}(1/t) = \frac{j_k(t) \cdot j_l(t)}{kl}$$

where $j_k(t) = \{t \bmod k\} \cdot k/1 \dots$ more precisely, $j_k(t)$ is the “fractional block index”: if $t \in [kn + j, kn + j + 1)$ then $j_k(t) = j \in \{0, \dots, k-1\}$.

The inner product is:

$$\langle \rho_{1/k}, \rho_{1/l} \rangle = \frac{1}{kl} \sum_{r=0}^{m-1} j_k(r) j_l(r) \cdot \sum_{n=0}^{\infty} \frac{1}{mn+r} - \frac{1}{mn+r+1}$$

where the inner sum is $(\psi((r+1)/m) - \psi(r/m))/m$.

Proof. By Lemma 1: $\rho_{1/k}(1/t) = j_k(t)/k$ and $\rho_{1/l}(1/t) = j_l(t)/l$ where $j_k(t)$ is the residue of $[t] \bmod k$. The function $j_k(t)j_l(t)/(kl)$ has period $m = \text{lcm}(k, l)$.

$$\langle \rho_{1/k}, \rho_{1/l} \rangle = \int_0^{\infty} \frac{j_k(t) j_l(t)}{kl} \frac{dt}{t^2} = \frac{1}{kl} \sum_{r=0}^{m-1} j_k(r) j_l(r) \sum_{n=0}^{\infty} \int_{mn+r}^{mn+r+1} \frac{dt}{t^2}.$$

The inner integral is $1/(mn+r) - 1/(mn+r+1) = 1/(mn+r)(mn+r+1) = (\psi((r+1)/m) - \psi(r/m))/m$. $\square$

Proposition 8 (Inner product $\langle \rho_{1/2}, \rho_{1/3} \rangle$). $m = \text{lcm}(2, 3) = 6$. The period-6 pattern of $j_2(r)j_3(r)$:

r	0	1	2	3	4	5
$j_2(r)$	0	1	0	1	0	1
$j_3(r)$	0	1	2	0	1	2
$j_2 j_3$	0	1	0	0	0	2

$$\langle \rho_{1/2}, \rho_{1/3} \rangle = \frac{1}{6} \left[1 \cdot \frac{\psi(2/6) - \psi(1/6)}{6} + 2 \cdot \frac{\psi(6/6) - \psi(5/6)}{6} \right] = \frac{6 \log 2 + 3 \log 3 - 2\pi/\sqrt{3}}{36}.$$

Numerically: $\approx (4.159 + 3.296 - 3.628)/36 \approx 0.106$.

Proof. From the table, only $r = 1$ (product 1) and $r = 5$ (product 2) contribute.

$$\langle \rho_{1/2}, \rho_{1/3} \rangle = \frac{1}{36} [(\psi(1/3) - \psi(1/6)) + 2(\psi(1) - \psi(5/6))].$$

From the Gauss digamma theorem: $\psi(1/3) - \psi(1/6) = 2 \log 2 + \pi/\sqrt{3}$ and $\psi(1) - \psi(5/6) = 2 \log 2 + \frac{3}{2} \log 3 - \frac{\pi\sqrt{3}}{2}$. Hence:

$$\langle \rho_{1/2}, \rho_{1/3} \rangle = \frac{2 \log 2 + \pi/\sqrt{3} + 4 \log 2 + 3 \log 3 - \pi\sqrt{3}}{36} = \frac{6 \log 2 + 3 \log 3 + \pi/\sqrt{3} - \pi\sqrt{3}}{36}.$$

$$\text{Since } \pi/\sqrt{3} - \pi\sqrt{3} = \pi(1/\sqrt{3} - \sqrt{3}) = -2\pi/\sqrt{3}: = \frac{6 \log 2 + 3 \log 3 - 2\pi/\sqrt{3}}{36}. \quad \square$$

5 The First Alignment Angle

Theorem 9 (First alignment angle). $\cos^2 \theta_2 = \log 2$.

Proof. Since $\rho_{1/1} = 0$, the space $\mathcal{V}_1 = \{0\}$, $f_1^* = 0$, $r_1 = 1_{(0,1)}$, and $v_2 = \rho_{1/2}$.

$$\cos^2 \theta_2 = \frac{|\langle 1_{(0,1)}, \rho_{1/2} \rangle|^2}{\|1_{(0,1)}\|^2 \|\rho_{1/2}\|^2} = \frac{(\log 2/2)^2}{1 \cdot (\log 2)/4} = \frac{(\log 2)^2/4}{(\log 2)/4} = \log 2. \quad \square$$

Corollary 10 (First entropy values).

$$d_1^2 = 1, \quad d_2^2 = 1 - \log 2 \approx 0.307, \quad c_2^* = \frac{\langle 1_{(0,1)}, \rho_{1/2} \rangle}{\|\rho_{1/2}\|^2} = \frac{(\log 2)/2}{(\log 2)/4} = 2.$$

The optimal approximation at step 2 is $f_2^ = 2\rho_{1/2}$. The residual is $r_2 = 1_{(0,1)} - 2\rho_{1/2}$.*

Theorem 11 (Second alignment angle).

$$\cos^2 \theta_3 = \frac{|\langle r_2, \rho_{1/3} \rangle|^2}{d_2^2 \|v_3\|^2}.$$

$$\begin{aligned} \text{The numerator: } \langle r_2, \rho_{1/3} \rangle &= \frac{\log 3}{3} - 2 \cdot \frac{6 \log 2 + 3 \log 3 - 2\pi/\sqrt{3}}{36} \\ &= \frac{\log 3}{3} - \frac{6 \log 2 + 3 \log 3 - 2\pi/\sqrt{3}}{18}. \end{aligned}$$

Numerically: $\approx 0.366 - 0.212 = 0.154$.

$$\begin{aligned} \text{The denominator factor: } \|v_3\|^2 &= \|\rho_{1/3}\|^2 - \frac{|\langle \rho_{1/3}, \rho_{1/2} \rangle|^2}{\|\rho_{1/2}\|^2} \\ &\approx 0.177 - (0.106)^2/0.173 \approx 0.177 - 0.065 = 0.112. \end{aligned}$$

$$\text{Therefore: } \cos^2 \theta_3 \approx \frac{(0.154)^2}{0.307 \times 0.112} \approx \frac{0.0237}{0.0344} \approx 0.689.$$

Remark 12 (Stability of the angle sequence). $\cos^2 \theta_2 = \log 2 \approx 0.693$ and $\cos^2 \theta_3 \approx 0.689$. Both are large and close to each other.

The partial product: $d_3^2 = d_1^2 \sin^2 \theta_2 \sin^2 \theta_3 \approx 1 \times (1 - \log 2)(1 - 0.689) \approx 0.307 \times 0.311 \approx 0.0955$.

The entropy decreases rapidly in the first few steps. This is consistent with RH: the BN flow approximates $1/\zeta$ well for small N because the first few BN functions capture the bulk of $1_{(0,1)}$. For large N , the angles must eventually decrease (Riemann–Lebesgue), but the early rapid convergence is arithmetic evidence of the structure of ζ .

6 Asymptotics

Theorem 13 (Norm asymptotics). *As $k \rightarrow \infty$:*

$$\|\rho_{1/k}\|^2 \sim \frac{C}{k}, \quad \text{where} \quad C = \int_0^1 x^2 \psi'(x) dx = \sum_{n=0}^{\infty} \int_0^1 \frac{x^2}{(x+n)^2} dx.$$

The constant is: $C = 1 + \sum_{n=1}^{\infty} (n+1 - 2n \log(1 + 1/n) - n^2/(n+1)) \approx 1.32$.

Proof. The Riemann sum approximation: $\sum_{j=1}^{k-1} j^2 (\psi((j+1)/k) - \psi(j/k)) \approx k^2 \int_0^1 x^2 \psi'(x) dx$ for large k (since the summand approximates $(j/k)^2 \psi'(j/k) \cdot k^2 \cdot (1/k)$). Therefore $\|\rho_{1/k}\|^2 \approx C/k$. $\square$

Remark 14 (Connection to the Mellin formula). The Mellin formula gives the same asymptotic: $\|\rho_{1/k}\|^2 \sim k^{-1} \cdot (2\pi)^{-1} \int_{-\infty}^{\infty} |\zeta(1/2 + it)|^2 / (1/4 + t^2) dt$. Comparing with the digamma formula, this identifies: $C = (2\pi)^{-1} \int_{-\infty}^{\infty} |\zeta(1/2 + it)|^2 / (1/4 + t^2) dt$. The digamma computation and the zeta integral give the same constant. The arithmetic of $\rho_{1/k}$ (digamma) and the analytic structure of ζ (Mellin) converge to the same number. They were always the same number.

References

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